

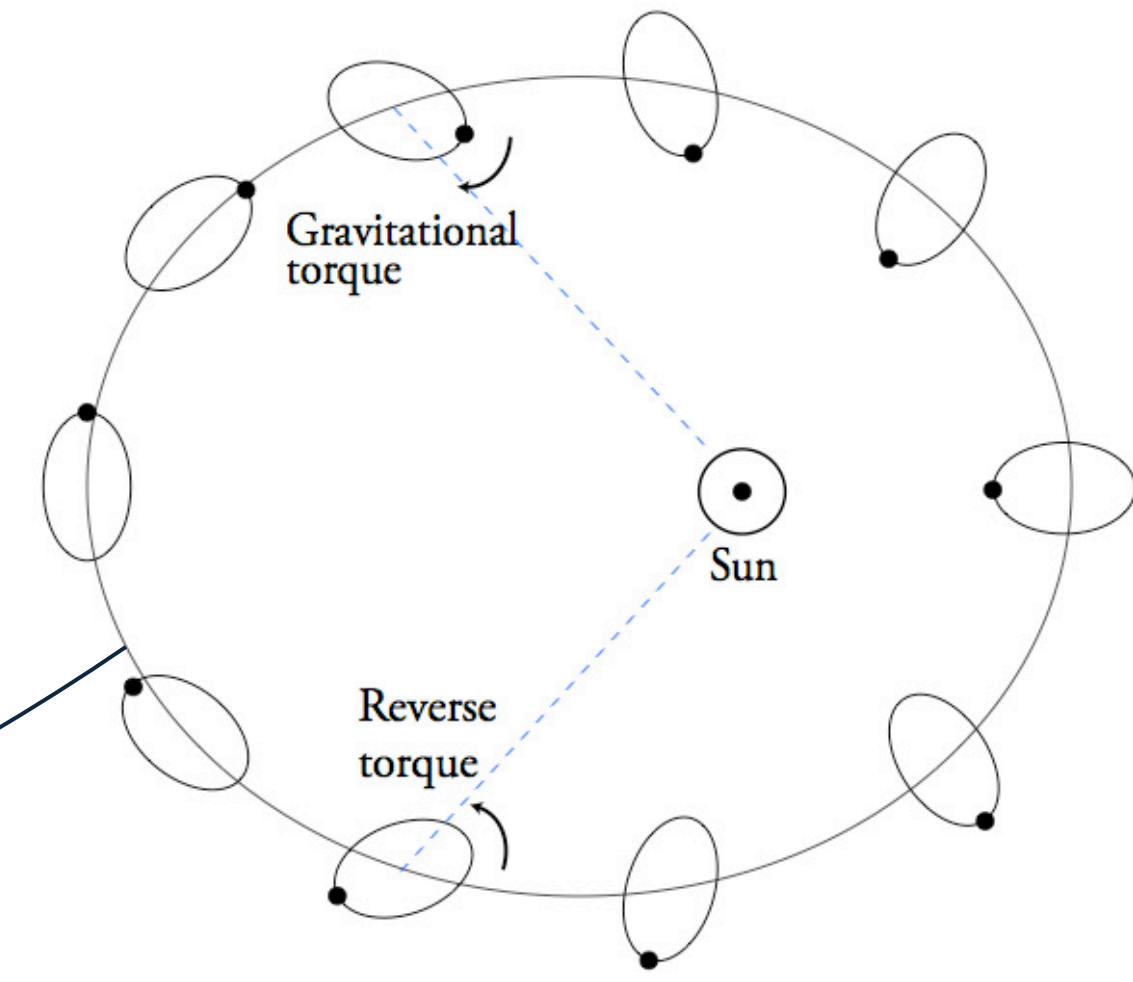
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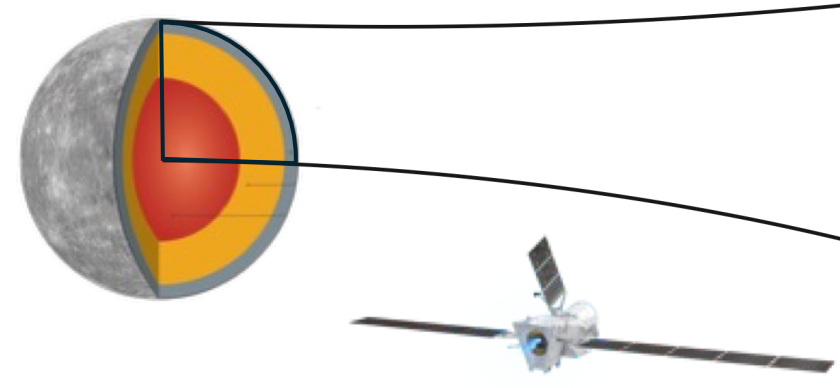
Scientific motivation

Mercury is locked in a 3:2 spin-orbit resonance. Because of its eccentric orbit and triaxial shape, the solar gravitational torque varies periodically, producing forced longitudinal librations.

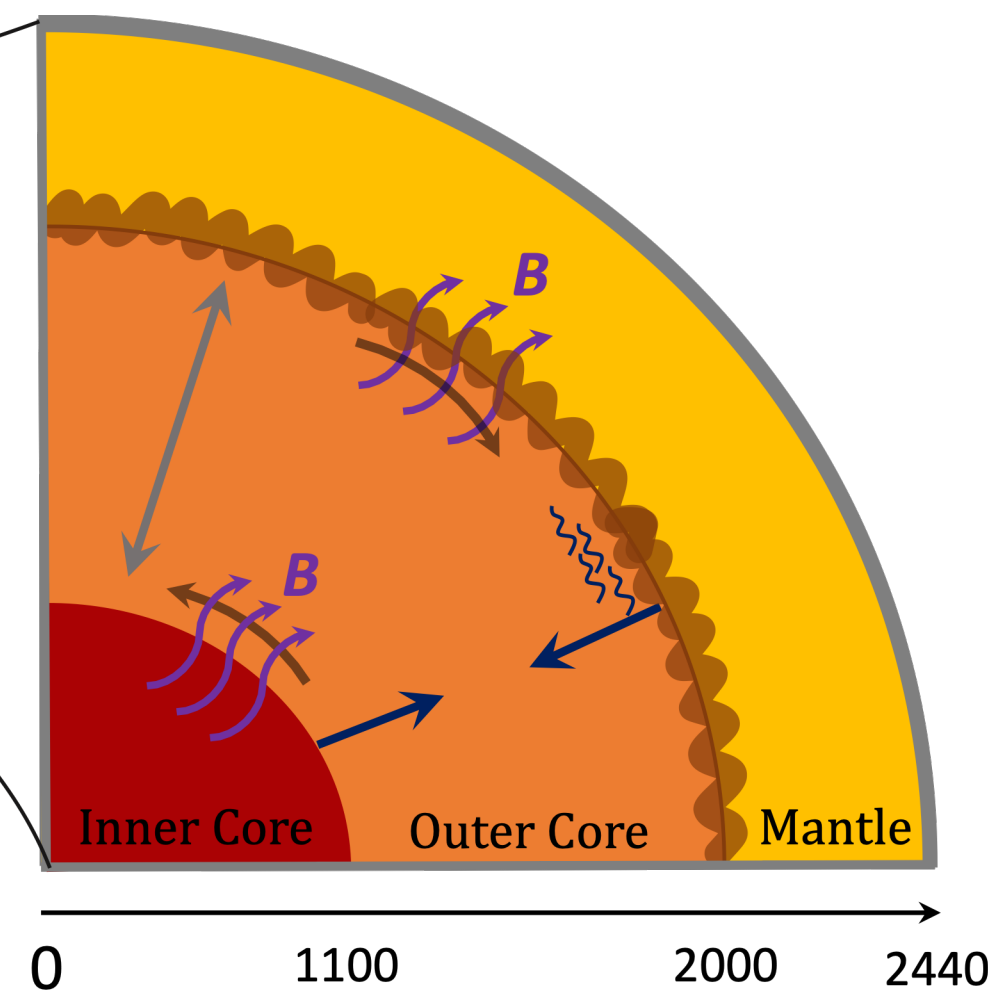
The amplitude and period of these librations depend on how angular momentum is exchanged between the mantle and the fluid core.



Interior structure



Pressure, viscous and electromagnetic coupling at the core-mantle boundary modify the rotational modes and therefore the observable librations.



Current models

Thin-wall approximation : Limited description of electromagnetic coupling

Objective

Three-layer model : Realistic electromagnetic coupling by solving magnetic diffusion throughout the mantle.

If the skin-depth

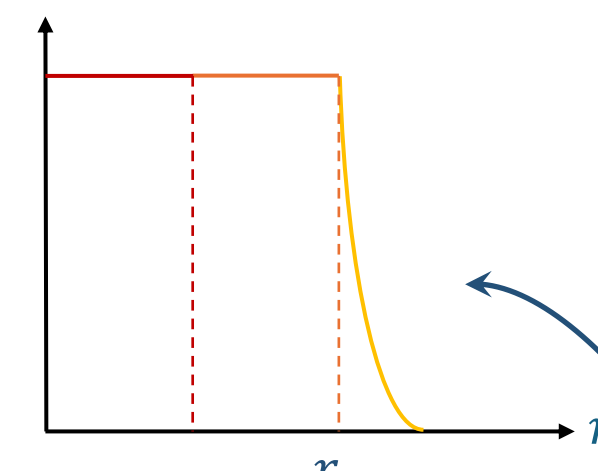
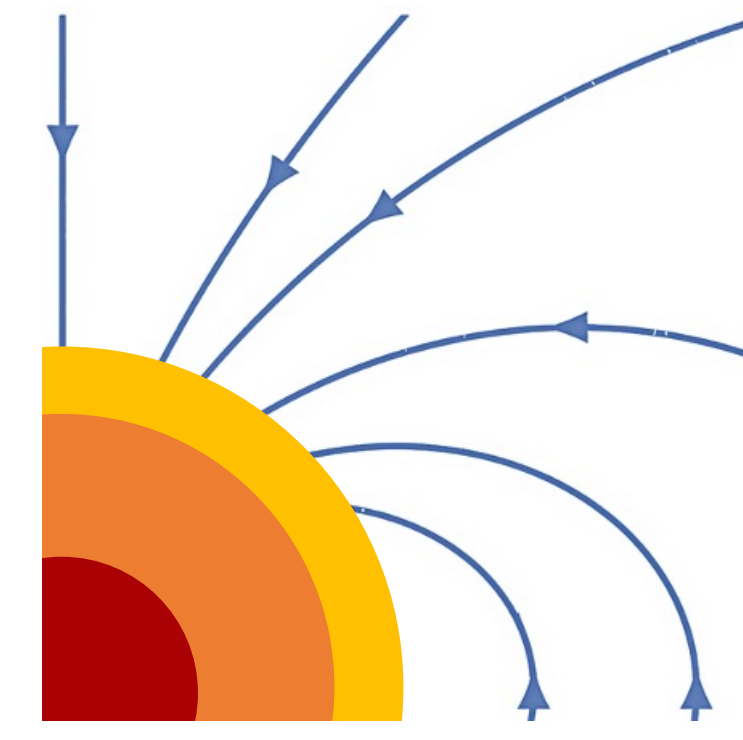
$$\delta_m = \sqrt{\frac{2}{\mu\sigma\omega}} \leq d$$

is smaller than the electrical conducting depth d of the mantle, the approximation is not valid anymore.

Scientific goal

Interpret future BepiColombo rotation measurements

Methods



KORE (Triana et al., 2021)

MHD in the fluid outer core

+

This work
Magnetic diffusion in the mantle
Variable conductivity profile $\sigma(r)$

Fully coupled 3-layer model

Inner core – Fluid outer core – Mantle

Induction equation under different forms

General Induction equation form

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{u} \times \mathbf{B}) - \nabla \times (\eta(r) \nabla \times \mathbf{B})$$

Inner core : no advection, η cst

$$\frac{\partial \mathbf{B}}{\partial t} = \eta \nabla^2 \mathbf{B} \rightarrow \text{Sph. Bessel's functions}$$

Fluid core : η cst

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{u} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}$$

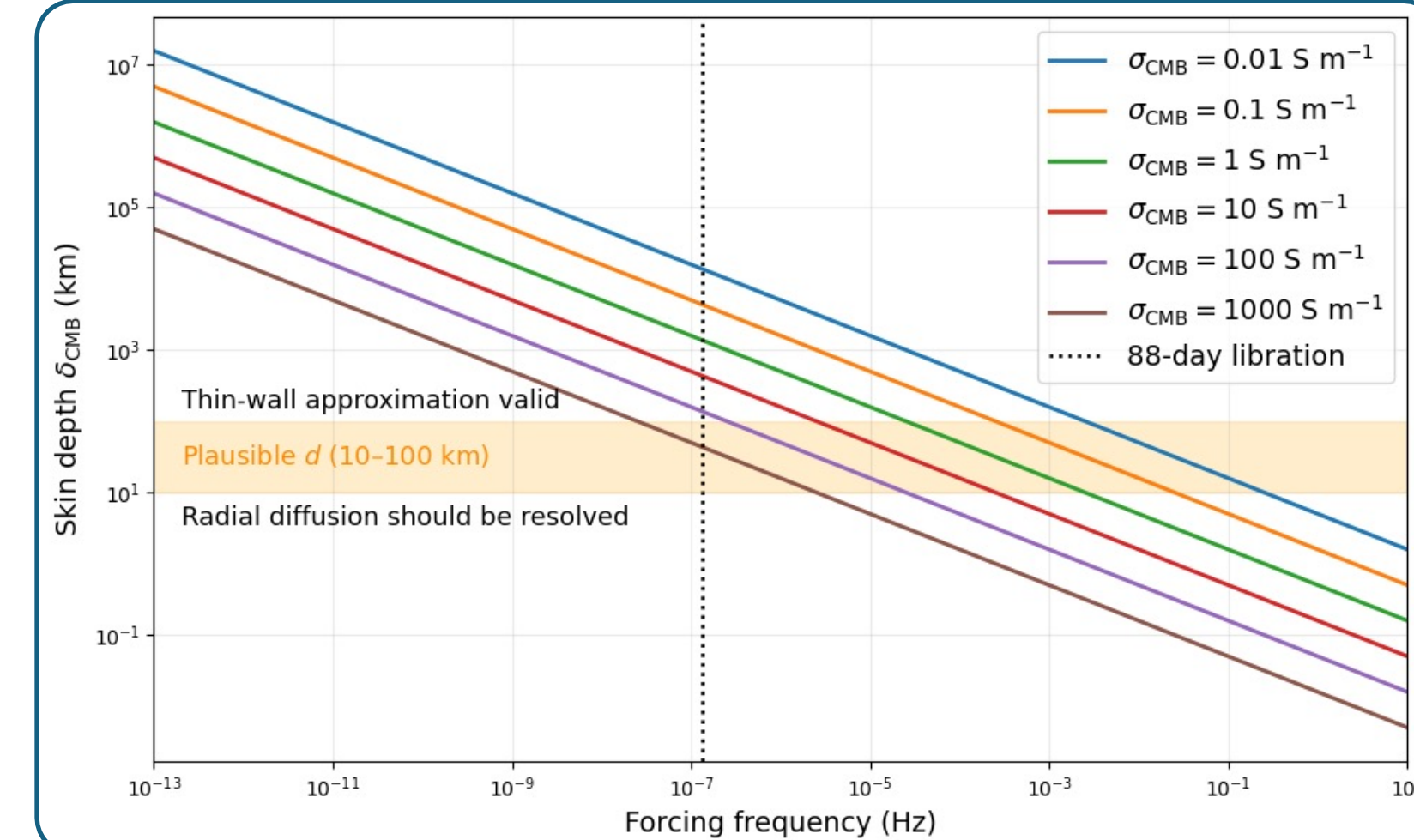
Mantle : no advection

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times (\eta(r) \nabla \times \mathbf{B})$$

With a variable diffusivity/conductivity profile

$$\sigma(r) = \sigma_{CMB} e^{a(r_{CMB}-r)} = \frac{1}{\mu_0 \eta(r)}$$

Thin-wall approximation or Mantle resolved ?



Development

Generalized Spherical Harmonics
Poloidal-Toroidal decomposition

Spectral discretization

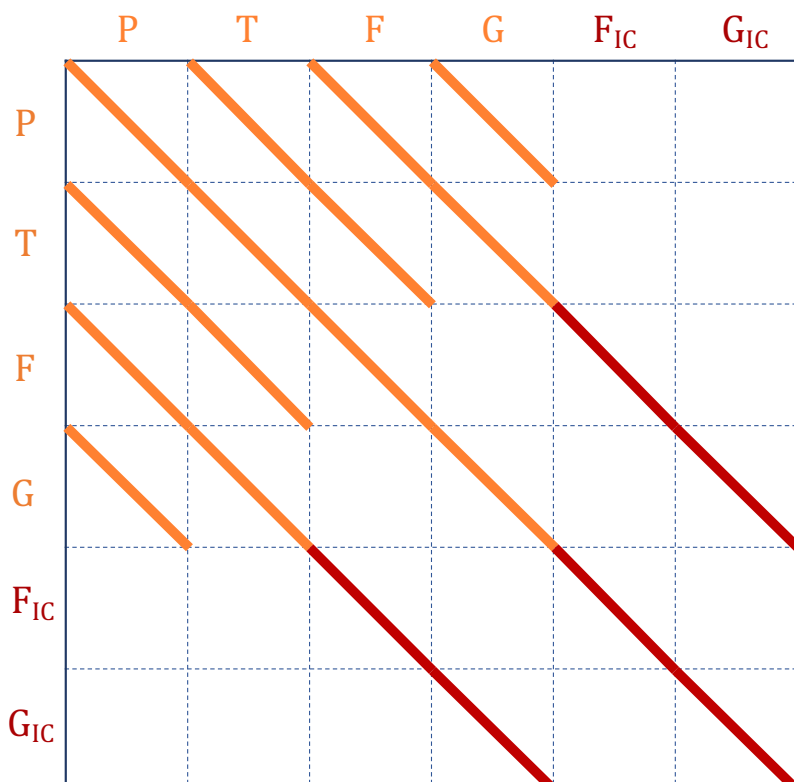
Ultraspherical Spectral Method
Chebychev Polynomial Basis
(Olver and Townsend, 2013)

Assembly

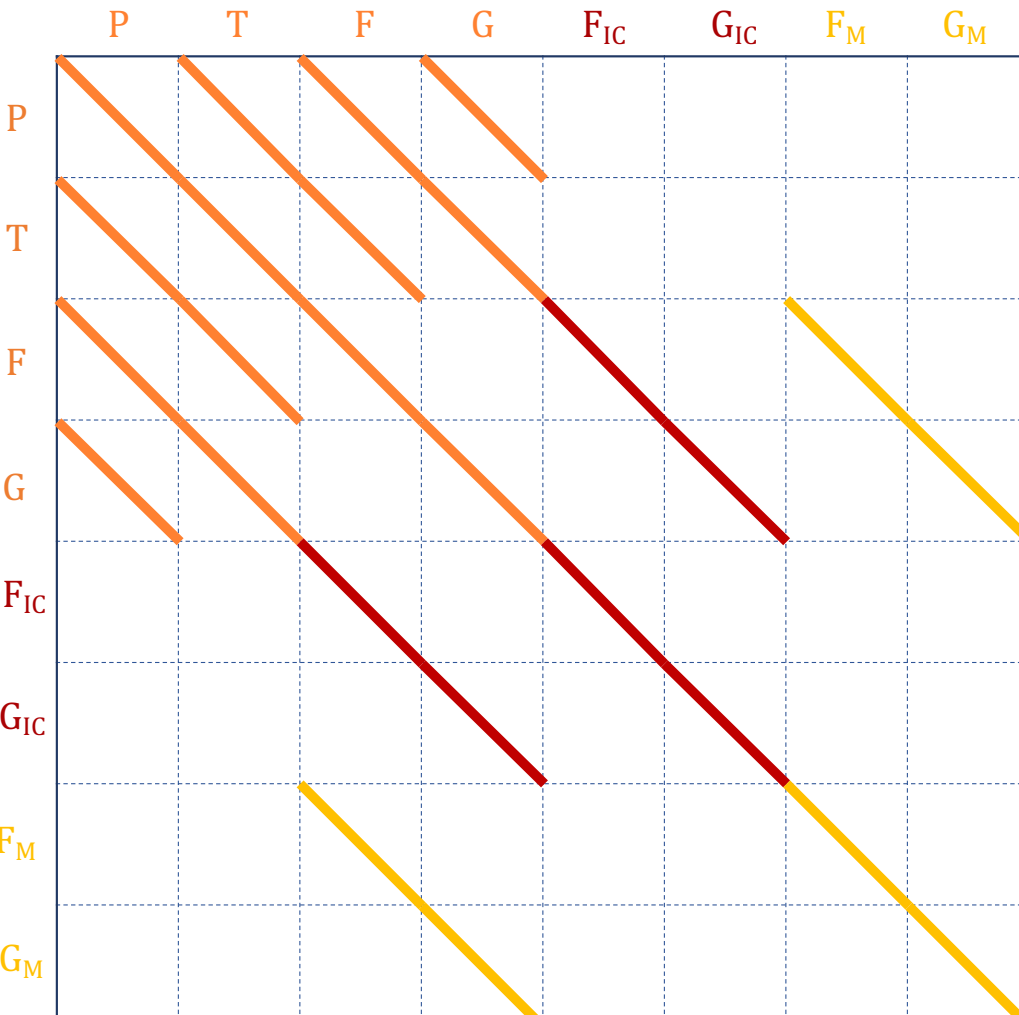
Matching conditions for $\mathbf{b}$ field from the ICB to the surface
Coupled eigenvalue / forced problem

Results

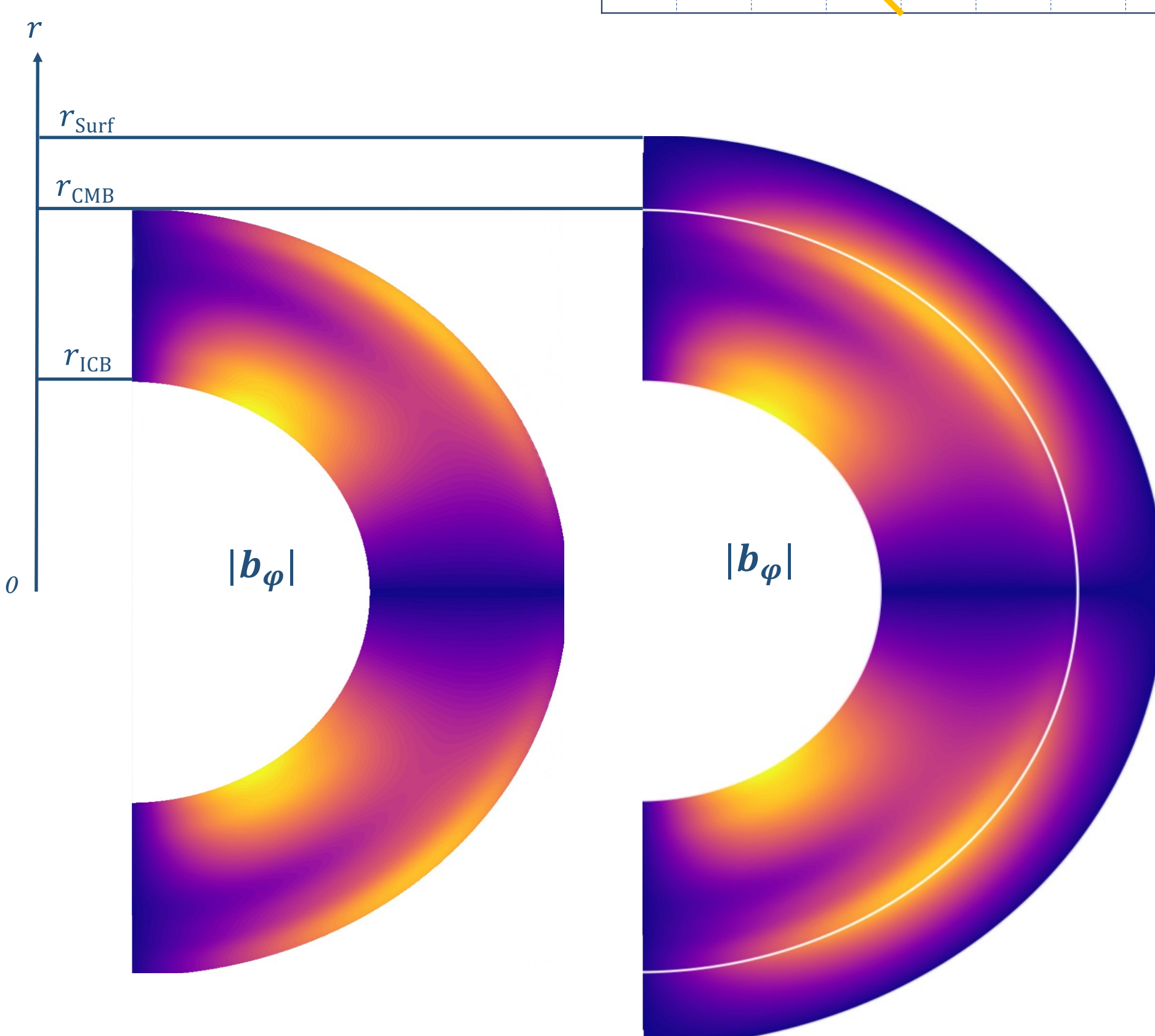
Kore - TWA



Kore - Mantle resolved



3rd layer



Solving the magnetic diffusion equation throughout the mantle with a realistic conductivity profile, enables a fully coupled three-layer model including the inner core, the fluid outer core and the mantle.

This new model aims to quantify how mantle conductivity modifies electromagnetic core-mantle coupling and the interpretation of the libration measurements that will be provided by BepiColombo.

References